# Problem-Solving, Ratios, Rates & Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

## Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

**1** [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{3}{x} = \frac{5}{20}$, what is $x$?

**2** [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity $f_0 = 1800$ increases by $20\%$ to $f_1$ and then decreases by $25\%$ to $f_2$. What is the final quantity $x = f_2$?

**3** [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{9}{2} = \frac{36}{x}$, what is the value of $x$?

**4** [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $6$ identical notebooks cost USD $66$ in total, what is the price per notebook?

**5** [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $3$ identical notebooks cost USD $18$ in total, what is the price per notebook?

**6** [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity of $390$ is decreased by $40\%$. What is the new value?

**7** [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity $p_0 = 3800$ increases by $10\%$ to $p_1$ and then decreases by $20\%$ to $p_2$. What is the final quantity $t = p_2$?

## Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

**8** [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{a}{b} = \frac{3}{5}$ and $b = 20$, what is $a$?

**9** [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{8}{10} = \frac{48}{x}$, what is the value of $x$?

**10** [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity of $70$ is increased by $50\%$. What is the new value?

© ClipSAT — clipsat.org | All Rights Reserved

**11** [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A car uses 8 liters of fuel to travel 100 km. How many liters to travel 350 km?

**12** [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A mixture is made of three ingredients in the ratio $3:1:1$. If the total mixture weighs $50$ grams, how many grams of the first ingredient are used?

**13** [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A mixture is made of three ingredients in the ratio $5:6:4$. If the total mixture weighs $60$ grams, how many grams of the first ingredient are used?

**14** [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A mixture is made of three ingredients in the ratio $3:1:3$. If the total mixture weighs $49$ grams, how many grams of the third ingredient are used?

## Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

**15** [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A car travels 240 miles in 4 hours. How far in 7 hours?

**16** [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What is $25\%$ of $420$?

**17** [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity of $40$ is decreased by $15\%$. What is the new value?

**18** [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A car travels at a constant speed of $30$ miles per hour for $7$ hours. How far does it travel?

**19** [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A line of best fit is $y = 8x + 40$ (hours studied vs. score). How many hours to reach a score of 88?

**20** [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

After a $40\%$ decrease, a price is USD $30$. What was the original price?

## Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

**21** [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A price decreases from USD 80 to USD 60. What is the percentage decrease?

**a.** 20%

**b.** 25%

**c.** 33%

**d.** 40%

**22** [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A recipe that serves 6 people calls for 10 cups of flour. How many cups of flour are needed to serve 19 people?

**a.** 95/3

**b.** 1903/60

**c.** 383/12

**d.** 377/12

**23** [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A store sells 4 apples for USD 3.00. At this rate, what is the cost of 12 apples?

**a.** USD 6.00

**b.** USD 8.00

**c.** USD 9.00

**d.** USD 12.00

**24** [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What is 35% of 200?

**a.** 60

**b.** 70

**c.** 80

**d.** 90

**25** [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A jacket originally priced at USD 60 is on sale for 20% off. What is the discount amount in dollars?

**a.** USD 10

**b.** USD 12

**c.** USD 15

**d.** USD 48

© ClipSAT — clipsat.org | All Rights Reserved

**26** [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A recipe requires 2 cups of sugar for every 3 cups of flour. If 9 cups of flour are used, how many cups of sugar are needed?

**a.** 4

**b.** 6

**c.** 8

**d.** 9

**27** [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A map scale indicates that 1 inch represents 25 miles. If two cities are 3.2 inches apart on the map, what is the actual distance between them?

**a.** 64 miles

**b.** 75 miles

**c.** 80 miles

**d.** 85 miles

**28** [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $x$ is directly proportional to $y$, and $x = 14$ when $y = 2$, what is $x$ when $y = 5$?

**a.** 25

**b.** 30

**c.** 35

**d.** 40

## Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

**29** [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

The scatter plot shows the relationship between study hours and scores. Which best describes the correlation?

**a.** Strong negative

**b.** Weak negative

**c.** No correlation

**d.** Strong positive

**30** [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A recipe that serves 6 people calls for 8 cups of flour. How many cups of flour are needed to serve 18 people?

**a.** 24

**b.** 14

**c.** 30

**d.** 31

© ClipSAT — clipsat.org | All Rights Reserved

**31** [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What is 15% of 80?

**a.** 8

**b.** 12

**c.** 15

**d.** 16

**32** [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A cyclist travels at a constant speed of 12 miles per hour. How many miles does the cyclist travel in 3.5 hours?

**a.** 36

**b.** 40

**c.** 42

**d.** 48

**33** [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A class of 30 students has 18 girls and 12 boys. What percentage of the class is boys?

**a.** 30%

**b.** 36%

**c.** 40%

**d.** 45%

**34** [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If a car uses 4 gallons of gasoline to travel 120 miles, how many gallons would it use to travel 300 miles at the same rate?

**a.** 8

**b.** 10

**c.** 12

**d.** 15

**35** [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If a 12-ounce beverage contains 150 calories, how many calories are in an 8-ounce serving of the same beverage?

**a.** 90

**b.** 100

**c.** 110

**d.** 120

© ClipSAT — clipsat.org | All Rights Reserved

**36** [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A car travels 180 miles in 3 hours. At this same rate, how far will it travel in 5 hours?

**a.** 240 miles

**b.** 270 miles

**c.** 300 miles

**d.** 360 miles

**37** [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

There are 1,000 milliliters in 1 liter. How many milliliters are in 3.4 liters?

**a.** 340 mL

**b.** 3,400 mL

**c.** 34,000 mL

**d.** 34 mL

## Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

**38** [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If 5 meters is approximately 16.4 feet, approximately how many feet are in 15 meters?

**a.** 32.8

**b.** 41.0

**c.** 49.2

**d.** 54.6

**39** [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

An item's price increased from USD 50 to USD 65. What was the percent increase?

**a.** 15%

**b.** 25%

**c.** 30%

**d.** 35%

**40** [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A runner completes a 5-kilometer race in 25 minutes. What is the runner's average speed in kilometers per hour?

**a.** 10 km/h

**b.** 12 km/h

**c.** 15 km/h

**d.** 20 km/h

© ClipSAT — clipsat.org | All Rights Reserved

**41** [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

An electronic device is discounted from USD 200 to USD 160. What is the percent discount?

**a.** 15%

**b.** 20%

**c.** 25%

**d.** 40%

**42** [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What number is 40% of 250?

**a.** 80

**b.** 90

**c.** 100

**d.** 120

**43** [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{a}{15} = \frac{4}{5}$, what is the value of $a$?

**a.** 10

**b.** 12

**c.** 14

**d.** 16

**44** [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A store sells notebooks in packs of 5 for USD 6. How much would 25 notebooks cost at this rate?

**a.** USD 24

**b.** USD 30

**c.** USD 35

**d.** USD 40

**45** [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What is 8% of USD 450?

**a.** USD 32

**b.** USD 36

**c.** USD 40

**d.** USD 45

© ClipSAT — clipsat.org | All Rights Reserved